

## Topic 2: Classes and Objects

### 2.1 Defining a Class

- A class definition starts with the keyword `class` followed by a name and a body in `{ }`
- Member variables (attributes) store the state of each object
- Member functions (methods) define the behaviour
- Access specifiers control visibility: `public` (accessible anywhere), `private` (only within the class)
- The class definition ends with a semicolon -- forgetting it is a common error

```
class Student {
private:
    string name;
    int    rollNo;
    double gpa;

public:
    void setData(string n, int r, double g);
    void display();
    double getGPA() { return gpa; } // inline definition
};
```

### 2.2 Implementing Member Functions Outside the Class

- Member functions can be defined outside the class body using the scope resolution operator `::`
- Syntax: `ReturnType ClassName::functionName(params) { ... }`
- Separating declaration (in `.h`) from definition (in `.cpp`) keeps code organised in larger projects

```
void Student::setData(string n, int r, double g) {
    name    = n;
    rollNo  = r;
    gpa     = g;
}

void Student::display() {
    cout << "Name: " << name
         << "   Roll: " << rollNo
         << "   GPA: "  << gpa << endl;
}
```

## 2.3 Creating and Using Objects

- Objects are created just like variables: `ClassName objectName;`
- Access public members with the dot operator: `obj.functionName()`
- Objects can also be created dynamically on the heap: `Student* s = new Student();`
- For pointer-based objects use arrow operator: `s->display(); delete s;`
- Stack objects are destroyed automatically when they go out of scope

```
int main() {  
    Student s1;                // stack object  
    s1.setData("Sara", 101, 3.9);  
    s1.display();  
  
    Student* s2 = new Student(); // heap object  
    s2->setData("Ahmed", 102, 3.5);  
    s2->display();  
    delete s2;  
    return 0;  
}
```

## 2.4 Array of Objects

- An array of objects stores multiple instances: `Student arr[3];`
  - Each element is a fully independent object
  - Access: `arr[0].display(); arr[1].setData(...);`
  - Useful for managing collections (e.g., all students in a class)
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